

Pre-analysis steps:

For our pre-analysis steps, we plan to first clean the data to make it easier for future exploratory data analysis. We also plan to filter our results to only include undergraduate students and clean column names:

- a. 'Timestamp': 'timestamp'
- b. 'Year at UVA': 'year'
- c. 'What is your Gender Identity?': 'gender'
- d. 'What Schools are you a Part of?': 'schools',
- e. 'Is a Hotdog a Sandwich?': 'is_sandwich',
- f. 'How strongly do you feel about your previous answer?': 'strength',
- g. 'How frequently do you eat hot dogs?': 'frequency'

Cleaning the data will improve the accuracy of our analysis and show a more accurate picture of the results we collected. For the pre-analysis, we also plan on conducting an exploratory analysis of our data. We will analyze the data to see if there are patterns within the opinion distribution of students saying “yes” or “no” to whether or not a hot dog is a sandwich. For example, we could examine to see if students in the School of Data Science are more likely to say “yes” to “Is a hot dog a sandwich?”. We can examine these patterns within all of these categories:

- a. Opinion distribution: % of yes/no
- b. Strength by opinions (1-5)
- c. Opinion by Year
- d. Frequency distribution
- e. Top schools (?)

Analysis methods:

H0: The proportion of all UVA undergraduate students who think a hot dog is a sandwich is $p=0.25$

Ha: $p \neq 0.25$

$n=175$

$x=71$

$\hat{p}=0.406$

1. We can calculate the sample proportion and perform a one-proportion z-test with the population proportion = 0.25.
2. By calculating the confidence interval (95% by default) from the actual proportion we got

Evaluation of success

To evaluate the success of the one-proportion z-test, we will determine if the resulting p-value is significant, comparing it to $\alpha = 0.05$. If the p-value is greater than 0.05, we fail to reject the null hypothesis, and we cannot say that the proportion of all UVA undergraduate students who think a hot dog is a sandwich is different than $\frac{1}{4}$. If the p-value is less than 0.05, then we have significant evidence to reject the null hypothesis and can conclude that the proportion is different than $\frac{1}{4}$.

From our actual data, we can get an interval [0.336, 0.48]. If the proportion in our null hypothesis fits in this interval, we fail to reject our null hypothesis, and vice versa.